SEÁN Ó MURCHÚ



Senior Software Engineer

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Backend engineer with eight years of experience designing distributed, event-driven systems. Specialises in functional programming, observability, and developer experience.

EXPERIENCE

Senior Software Engineer
Jun 2022 – Present

Acme Corp · Dublin, Ireland

Platform team lead. Owns the event-sourcing stack underpinning the wider product surface.

- Led the migration of a customer-facing monolith into a set of event-driven microservices, halving p99 latency.
- Designed and rolled out an event-sourcing platform now used by four product teams.
- Mentored three engineers from mid to senior level.

Software Engineer
Jun 2019 – Jun 2021

Liffey Labs · Remote

- Built and shipped the first version of a SaaS product from scratch alongside a two-person team.
- Introduced the CI/CD pipeline and developer-onboarding flow that scaled the engineering org from 3 to 15.

VOLUNTEER

Mentor
Jun 2020 – Present

CoderDojo Dublin

Volunteer-led coding club introducing young people to programming.

- Weekly mentoring sessions for 10–14 year-olds learning to code.
- Curate the Scala / functional-programming track.

PROJECTS

Application

Hyperion
Jan 2024 – Aug 2024

Acme Corp · Lead, Architect

Distributed task scheduler in Rust

- Handled 10k tasks/s on a single node.
- Designed an idempotent retry protocol.

Rust · Tokio · gRPC

Crucible
Sep 2024 – Present

Maintainer

Migration tool for legacy databases

- Schema diffing across PostgreSQL and MySQL.

AREAS OF FOCUS

- Distributed systems: Kafka, CDC, idempotent consumers, event sourcing.
- Functional programming and type-driven design in Scala and Haskell.
- Developer experience: tooling, CI/CD, observability.
- Engineering culture: ADRs, RFCs, mentoring.

SKILLS

Languages – Scala · Haskell · Python · Go · Rust

Infra – Kafka · AWS · Terraform · Docker · Kubernetes

LANGUAGES

English ●●●●●●

Irish ●●●◐○○

French ●●●●○○

Portuguese ●●◐○○○

EDUCATION

M.Sc. in Computer Science
2017 – 2019

Tallaght Institute of Technology

Distinction

Distributed Systems · Type Theory · Concurrency

Computer Science
2014 – 2017

Trinity College Dublin

Library
Tinkerbell

Tiny IRC bot

Lua

Other
Quill

PUBLICATIONS

Articles
- *Event Sourcing in Practice*.
  Personal Blog · 14 Jun 2024

  A walk-through of idempotent Kafka consumers and the replay strategy that ships with them.
- *Untyped Note*.
  Jan 2024

Conference Papers
- *Idempotent Kafka Consumers at Scale*.
  EuroSys · Apr 2024

Talks
- *Designing for Failure*.
  ScalaIO · Sep 2023

Books
- *Functional Domain Modelling*.
  Self-published · 2022

INTERESTS

Music: Trad, Jazz
Sport: Hurling, Climbing
Travel: Japan, Iceland

CERTIFICATIONS

Certified Kubernetes Administrator — Aug 2023
CNCF

Certified Kubernetes Application Developer — Jan 2024
CNCF

AWS Solutions Architect — Professional — May 2023
AWS

AWS DevOps Engineer — Professional — Sep 2024
AWS

Hashicorp Terraform Associate — Dec 2022
Hashicorp

Certified Kubernetes Security Specialist — Mar 2024
Linux Foundation

Google Cloud Professional Architect — Nov 2023
Google Cloud

AWARDS

Best Paper Award — Sep 2024
ScalaConf
For Idempotent Kafka Consumers.

Dean's List

REFERENCES

Dr Ada Lovelace
Seán is a calm, methodical engineer who left every system he touched in a better state than he found it. Recommended without reservation.

Pragmatic, curious, unfailingly kind — would hire again in a heartbeat.